

Домашнее задание 3.

Автоматизированное развертывание Nginx + Apache при помощи СУК Ansible

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Подготовка к выполнению задания

Для данного задания я создал две виртуальные машины на Amazon AWS EC2 (eu-central-1): одну для Apache, другую для Nginx.

Настройка созданных виртуальных машин на Amazon Aws

The screenshot displays the AWS Management Console for the EC2 service. The left-hand navigation pane includes sections for EC2 (Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager), Images (AMIs, AMI Catalog), Elastic Block Store (Volumes, Snapshots, Lifecycle Manager), and Network & Security. The main area shows the 'Instances (1/2)' page. A table lists two instances: 'my-linux-lab' (ID: i-098e7bb7fe5fba5c0) and 'my-linux-lab-a...' (ID: i-0694db5887833cb9d). The second instance is selected, and its details are shown below. The details include an 'Instance summary' with fields for Instance ID, IPv6 address, Hostname type, Public IPv4 address, Instance state (Running), Private IPv4 addresses, Public DNS, and Private IP DNS name (IPv4 only).

Настройка Security Groups и открытие портов

Порты открыл для всех IP-адресов для удобства, в production вариант так делать, конечно же не буду.

The screenshot displays the AWS Management Console for the Security Groups service. The left-hand navigation pane includes sections for Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, Elastic Block Store, Network & Security, and Load Balancing. The main area shows the 'Security Groups (1/2)' page. A table lists two security groups: 'sg-0916c773e8ad727f4' (default) and 'sg-0282d63241eac9eb5' (launch-wizard-1). The second security group is selected, and its details are shown below. The details include tabs for Details, Inbound rules, Outbound rules, Sharing, VPC associations, and Tags. The 'Inbound rules' tab is active, showing a table with two rules: one for HTTP (TCP) and one for SSH (TCP), both allowing traffic from all IP addresses.

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-0042869876041bf00	HTTP	TCP	80	Cus... 0.0.0.0/0		Delete
sgr-061cacadc6108f711	SSH	TCP	22	Cus... 0.0.0.0/0		Delete

[Add rule](#)

⚠ Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

Настройка ~/.ssh/config

Для удобства настройки на своей виртуальной машине Linux ubuntu создал след. файл:

```
Host my-aws-linux-nginx
  HostName 18.196.82.66
  User ubuntu
  IdentityFile ~/.ssh/my-key-pair.pem
  IdentitiesOnly yes

Host my-aws-linux-apache
  HostName 3.70.236.149
  User ec2-user
  IdentityFile ~/.ssh/my-key-pair.pem
  IdentitiesOnly yes
```

Проверил, что работает доступ по SSH с моей виртуальной машины до серверов на AWS

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ ssh my-aws-linux-nginx
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Dec 1 19:27:53 UTC 2025

System load: 0.24           Temperature: -273.1 C
Usage of /: 43.3% of 6.71GB Processes: 115
Memory usage: 26%          Users logged in: 0
Swap usage: 0%             IPv4 address for ens5: 172.31.27.76

 * Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.

  https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Dec 1 19:11:49 2025 from 81.16.6.137
ubuntu@ip-172-31-27-76:~$
```

```

#_
~\_##### Amazon Linux 2023
~~~\#####
~~~\###|
~~~\#| https://aws.amazon.com/linux/amazon-linux-2023
~~~V~'~>
~~~~
~~~~~*~
~~~~~\|_
~~~~~\m|'
Last login: Mon Dec 1 19:12:07 2025 from 81.16.6.137
[ec2-user@ip-172-31-23-221 ~]$

```

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ ansible -i inventory.ini apache -m ping
my-aws-linux-apache | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

Подготовка структуры папок

Nginx

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ mkdir -p ~/ansible-webstack
cd ~/ansible-webstack

mkdir -p roles/nginx/{tasks,templates,handlers}
touch inventory.ini nginx.yml \
    roles/nginx/tasks/main.yml \
    roles/nginx/handlers/main.yml \
    roles/nginx/templates/index.html.j2
```

Apache

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ mkdir -p roles/apache/{tasks,templates,handlers}
touch roles/apache/tasks/main.yml \
    roles/apache/handlers/main.yml \
    roles/apache/templates/index.html.j2
```

Создание роли Ansible для установки и настройки Anginx и Apache

Nginx

Создаем необходимые файлы

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ nano roles/nginx/tasks/main.yml
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ nano roles/nginx/handlers/main.yml
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ nano roles/nginx/t
tasks/      templates/
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ nano roles/nginx/t
tasks/      templates/
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ nano roles/nginx/templates/index.html.j2
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ nano nginx.yml
```

tasks/main.yml

```
GNU nano 7.2                                     roles/nginx/tasks/main.yml
---
- name: Install Nginx
  ansible.builtin.package:
    name: nginx
    state: present
  notify: Restart nginx

- name: Ensure Nginx service is enabled and running
  ansible.builtin.service:
    name: nginx
    state: started
    enabled: true

- name: Deploy custom index.html to Nginx default site
  ansible.builtin.template:
    src: index.html.j2
    dest: /var/www/html/index.html
    owner: www-data
    group: www-data
    mode: '0644'
```

handlers/main.yml

```
GNU nano 7.2                                     roles/nginx/handlers/main.yml
---
- name: Restart nginx
  ansible.builtin.service:
```

```
name: nginx
state: restarted
```

templates/index.html

```
GNU nano 7.2                                     roles/nginx/templates/index.html.j2
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Nginx via Ansible</title>
</head>
<body>
  <h1>Nginx is working! </h1>

  <h2>Deployment info</h2>
  <ul>
    <li><strong>Ansible inventory name:</strong> {{ inventory_hostname }}</li>
    <li><strong>System hostname:</strong> {{ ansible_hostname }}</li>
    <li><strong>FQDN:</strong> {{ ansible_fqdn }} </li>
    <li><strong>Primary IP address:</strong> {{ ansible_default_ipv4.address }}</li>
  </ul>

  <h2>System info</h2>
  <ul>
    <li><strong>OS:</strong> {{ ansible_distribution }} {{ ansible_distribution_version
  }}</li>
    <li><strong>Total RAM:</strong> {{ ansible_memtotal_mb }} MB</li>
    <li><strong>vCPUs:</strong> {{ ansible_processor_vcpus }}</li>
    <li><strong>CPU model:</strong> {{ ansible_processor[0] }}</li>
    <li><strong>Vendor / platform:</strong> {{ ansible_system_vendor }} {{
ansible_product_name }}</li>
  </ul>

  <p>This page was deployed by Ansible role <strong>nginx</strong>.</p>
  <p>Author - Vladimir Kuryndin </p>
</body>
</html>
```

Apache

tasks/main.yml

```
GNU nano 7.2                                     roles/apache/tasks/main.yml
---
- name: Install Apache (httpd)
  ansible.builtin.package:
    name: httpd
    state: present
  notify: Restart apache

- name: Ensure Apache (httpd) service is enabled and running
  ansible.builtin.service:
    name: httpd
    state: started
    enabled: true

- name: Deploy custom index.html to Apache default site
  ansible.builtin.template:
    src: index.html.j2
    dest: /var/www/html/index.html
    owner: apache
    group: apache
    mode: '0644'
```

handlers/main.yml

```
GNU nano 7.2 roles/apache/handlers/main.yml
---
- name: Restart apache
  ansible.builtin.service:
    name: httpd
    state: restarted
```

templates/index.html

```
GNU nano 7.2 roles/apache/templates/index.html.j2
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Apache via Ansible</title>
</head>
<body>
  <h1>Apache (httpd) is working! </h1>

  <h2>Deployment info</h2>
  <ul>
    <li><strong>Inventory name:</strong> {{ inventory_hostname }}</li>
    <li><strong>System hostname:</strong> {{ ansible_hostname }}</li>
    <li><strong>FQDN:</strong> {{ ansible_fqdn }} </li>
    <li><strong>Primary IP address:</strong> {{ ansible_default_ipv4.address }}</li>
  </ul>

  <h2>System info</h2>
  <ul>
    <li><strong>OS:</strong> {{ ansible_distribution }} {{ ansible_distribution_version }}</li>
    <li><strong>Total RAM:</strong> {{ ansible_memtotal_mb }} MB</li>
    <li><strong>vCPUs:</strong> {{ ansible_processor_vcpus }}</li>
    <li><strong>CPU model:</strong> {{ ansible_processor[0] }}</li>
    <li><strong>Vendor / platform:</strong> {{ ansible_system_vendor }} {{ ansible_product_name }}</li>
  </ul>

  <p>This page was deployed by Ansible role <strong>apache</strong> on Amazon Linux.</p>
  <p>Created by Vladimir Kuryndin.</p>
</body>
</html>
```

Плейбук, который использует созданные роли для развертывания Nginx и Apache на удаленных серверах

Nginx

```
GNU nano 7.2 nginx.yml
---
- name: Configure Nginx on nginx group
  hosts: nginx
  become: true
  gather_facts: true
```



```
roles:
  - nginx
```

Apache

```
GNU nano 7.2                                     apache.yml
---
- name: Configure Apache on apache group
  hosts: apache
  become: true
  gather_facts: true

roles:
  - apache
```

Общий

Также я создал общий единый плейбук: который запускаем оба сервера сразу. Он запускает роль nginx для группы [nginx] и роль apache для группы [apache].

```
GNU nano 7.2                                     site.yml
---
- name: Configure Nginx on nginx group
  hosts: nginx
  become: true
  gather_facts: true

roles:
  - nginx

- name: Configure Apache on apache group
  hosts: apache
  become: true
  gather_facts: true

roles:
  - apache
```

Выполнение плейбука

Nginx

```
ansible-playbook -i inventory.ini nginx.yml
```

```

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ ansible-playbook -i inventory.ini nginx.yml

PLAY [Configure Nginx on nginx group] *****

TASK [Gathering Facts] *****
ok: [my-aws-linux-nginx]

TASK [nginx : Install Nginx] *****
ok: [my-aws-linux-nginx]

TASK [nginx : Ensure Nginx service is enabled and running] *****
ok: [my-aws-linux-nginx]

TASK [nginx : Deploy custom index.html to Nginx default site] *****
changed: [my-aws-linux-nginx]

PLAY RECAP *****
my-aws-linux-nginx      : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

Apache

```
ansible-playbook -i inventory.ini apache.yml
```

```

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ ansible-playbook -i inventory.ini apache.yml

PLAY [Configure Apache on apache group] *****

TASK [Gathering Facts] *****
ok: [my-aws-linux-apache]

TASK [apache : Install Apache (httpd)] *****
changed: [my-aws-linux-apache]

TASK [apache : Ensure Apache (httpd) service is enabled and running] *****
changed: [my-aws-linux-apache]

TASK [apache : Deploy custom index.html to Apache default site] *****
changed: [my-aws-linux-apache]

RUNNING HANDLER [apache : Restart apache] *****
changed: [my-aws-linux-apache]

PLAY RECAP *****
my-aws-linux-apache    : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

Общий

```
ansible-playbook -i inventory.ini site.yml
```

```

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ ansible-playbook -i inventory.ini site.yml

PLAY [Configure Nginx on nginx group] *****

TASK [Gathering Facts] *****
ok: [my-aws-linux-nginx]

TASK [nginx : Install Nginx] *****
ok: [my-aws-linux-nginx]

TASK [nginx : Ensure Nginx service is enabled and running] *****
ok: [my-aws-linux-nginx]

TASK [nginx : Deploy custom index.html to Nginx default site] *****
ok: [my-aws-linux-nginx]

PLAY [Configure Apache on apache group] *****

TASK [Gathering Facts] *****
ok: [my-aws-linux-apache]

TASK [apache : Install Apache (httpd)] *****
ok: [my-aws-linux-apache]

TASK [apache : Ensure Apache (httpd) service is enabled and running] *****
ok: [my-aws-linux-apache]

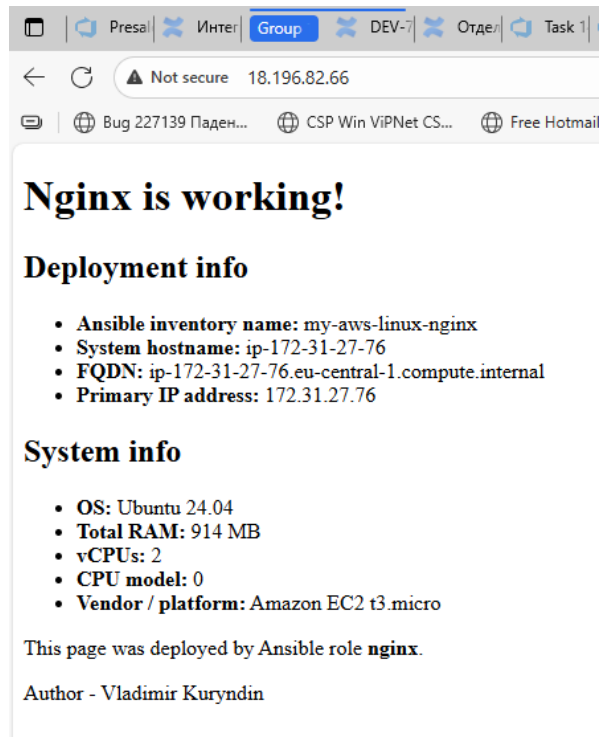
TASK [apache : Deploy custom index.html to Apache default site] *****
ok: [my-aws-linux-apache]

PLAY RECAP *****
my-aws-linux-apache    : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
my-aws-linux-nginx     : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

Проверка работоспособности развернутых сервисов с помощью браузера или командной строки

Nginx



The screenshot shows a web browser window with the address bar displaying "18.196.82.66". The page content includes a large heading "Nginx is working!", a section titled "Deployment info" with a bulleted list of configuration details, and a section titled "System info" with another bulleted list of hardware and software details. At the bottom, it states "This page was deployed by Ansible role nginx." and "Author - Vladimir Kuryndin".

Nginx is working!

Deployment info

- **Ansible inventory name:** my-aws-linux-nginx
- **System hostname:** ip-172-31-27-76
- **FQDN:** ip-172-31-27-76.eu-central-1.compute.internal
- **Primary IP address:** 172.31.27.76

System info

- **OS:** Ubuntu 24.04
- **Total RAM:** 914 MB
- **vCPUs:** 2
- **CPU model:** 0
- **Vendor / platform:** Amazon EC2 t3.micro

This page was deployed by Ansible role **nginx**.

Author - Vladimir Kuryndin

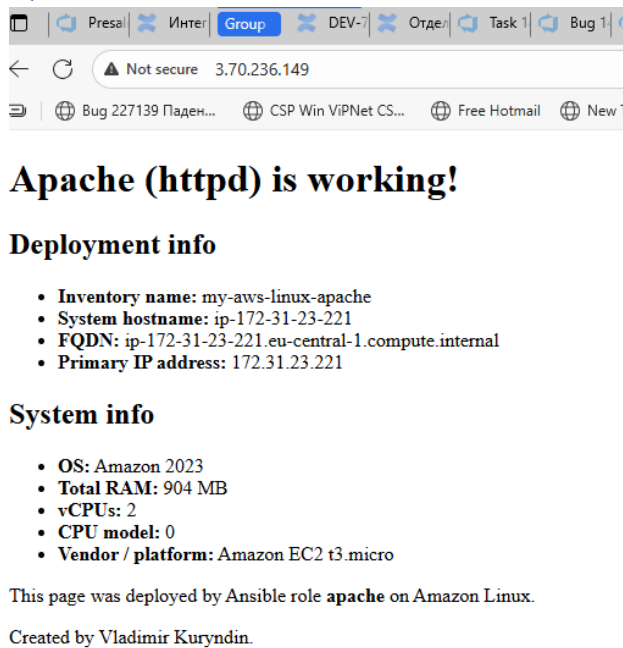
```
vladimir.kuryndin@vladimir-kuryndin-Virtual-Platform:~/ansible-webstack$ curl http://18.196.82.66
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Nginx via Ansible</title>
</head>
<body>
  <h1>Nginx is working! </h1>

  <h2>Deployment info</h2>
  <ul>
    <li><strong>Ansible inventory name:</strong> my-aws-linux-nginx</li>
    <li><strong>System hostname:</strong> ip-172-31-27-76</li>
    <li><strong>FQDN:</strong> ip-172-31-27-76.eu-central-1.compute.internal </li>
    <li><strong>Primary IP address:</strong> 172.31.27.76</li>
  </ul>

  <h2>System info</h2>
  <ul>
    <li><strong>OS:</strong> Ubuntu 24.04</li>
    <li><strong>Total RAM:</strong> 914 Mb</li>
    <li><strong>vCPUs:</strong> 2</li>
    <li><strong>CPU model:</strong> 0</li>
    <li><strong>Vendor / platform:</strong> Amazon EC2 t3.micro</li>
  </ul>

  <p>This page was deployed by Ansible role <strong>nginx</strong>.</p>
  <p>Author - Vladimir Kuryndin </p>
</body>
</html>
```

Apache



The screenshot shows a web browser window with the address bar displaying "3.70.236.149". The page content includes a large heading "Apache (httpd) is working!", a section titled "Deployment info" with a bulleted list of configuration details, and a section titled "System info" with another bulleted list of hardware and software details. At the bottom, it states "This page was deployed by Ansible role apache on Amazon Linux." and "Created by Vladimir Kuryndin".

Apache (httpd) is working!

Deployment info

- **Inventory name:** my-aws-linux-apache
- **System hostname:** ip-172-31-23-221
- **FQDN:** ip-172-31-23-221.eu-central-1.compute.internal
- **Primary IP address:** 172.31.23.221

System info

- **OS:** Amazon 2023
- **Total RAM:** 904 MB
- **vCPUs:** 2
- **CPU model:** 0
- **Vendor / platform:** Amazon EC2 t3.micro

This page was deployed by Ansible role **apache** on Amazon Linux.

Created by Vladimir Kuryndin.

```
Vladimir.kuryndin@vladimir-kuryndin-Virtual-Platform:~/ansible-webstack$ curl http://3.70.236.149
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Apache via Ansible</title>
</head>
<body>
  <h1>Apache (httpd) is working! </h1>

  <h2>Deployment info</h2>
  <ul>
    <li><strong>Inventory name:</strong> my-aws-linux-apache</li>
    <li><strong>System hostname:</strong> ip-172-31-23-221</li>
    <li><strong>FQDN:</strong> ip-172-31-23-221.eu-central-1.compute.internal </li>
    <li><strong>Primary IP address:</strong> 172.31.23.221</li>
  </ul>

  <h2>System info</h2>
  <ul>
    <li><strong>OS:</strong> Amazon 2023</li>
    <li><strong>Total RAM:</strong> 904 MB</li>
    <li><strong>vCPUs:</strong> 2</li>
    <li><strong>CPU model:</strong> 0</li>
    <li><strong>Vendor / platform:</strong> Amazon EC2 t3.micro</li>
  </ul>

  <p>This page was deployed by Ansible role <strong>apache</strong> on Amazon Linux.</p>
  <p>Created by Vladimir Kuryndin.</p>
</body>
</html>
```

Удаление Nginx и Apache

Я создал общий плейбук для удаления всех серверов сразу. По отдельности для удаления плейбуки также доступны.

```
GNU nano 7.2                                                                    site-
remove.yml
---
- name: Remove Nginx from nginx group
  hosts: nginx
  become: true
  gather_facts: false

  tasks:
    - name: Stop and disable Nginx service
      ansible.builtin.service:
        name: nginx
        state: stopped
        enabled: false

    - name: Remove Nginx package
      ansible.builtin.package:
        name: nginx
        state: absent

    - name: Remove Nginx index.html (if still exists)
      ansible.builtin.file:
        path: /var/www/html/index.html
        state: absent

- name: Remove Apache (httpd) from apache group
  hosts: apache
  become: true
  gather_facts: false

  tasks:
    - name: Stop and disable Apache (httpd) service
      ansible.builtin.service:
        name: httpd
        state: stopped
        enabled: false

    - name: Remove Apache (httpd) package
      ansible.builtin.package:
        name: httpd
        state: absent

    - name: Remove Apache index.html (if still exists)
      ansible.builtin.file:
        path: /var/www/html/index.html
        state: absent
```

Общая структура всех итоговых файлов

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~/ansible-webstack$ tree ~/ansible-webstack
/home/vladimir-kuryndin/ansible-webstack
├── apache-remove.yml
├── apache.yml
├── inventory.ini
├── nginx-remove.yml
├── nginx.yml
├── roles
│   ├── apache
│   │   ├── handlers
│   │   │   └── main.yml
│   │   ├── tasks
│   │   │   └── main.yml
│   │   └── templates
│   │       └── index.html.j2
│   └── nginx
│       ├── handlers
│       │   └── main.yml
│       ├── tasks
│       │   └── main.yml
│       └── templates
│           └── index.html.j2
├── site-remove.yml
└── site.yml

10 directories, 13 files
```

Дополнительная информация

Используемые адреса для проверки результатов: [Apache via Ansible](#) , [Nginx via Ansible](#)

.Виртуальные машины будут доступны ограниченное время после сдачи задания.